

Matthieu Blanke

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RESEARCH POSITIONS

NYU Courant Institute of Mathematical Sciences **New York, USA**
Postdoctoral Associate **2025**
Deep learning for climate, advised by Sara Shamekh and Pierre Gentine.

EDUCATION

Inria Paris and DI ENS **Paris, France**
PhD **2021-2024**

Deep learning for physical systems, advised by Marc Lelarge.

École Normal Supérieure Paris-Saclay **Paris, France**
Master of Science *Mathématiques, Vision, Apprentissage (MVA)* **2020-2021**
Machine learning, optimization and statistics. Also passed exams of statistical physics courses at ENS Paris.

École polytechnique **Paris, France**
Engineer's degree *Cycle ingénieur polytechnicien* **2017 - 2021**
Applied mathematics, computer science, theoretical physics

WORK EXPERIENCE

Inria Paris **Paris, France**
Research internship **April - September 2021**
Deep implicit layers with applications to physical systems. Advised by Marc Lelarge.

Econophysix **Paris, France**
Research internship **June - September 2020**
Particle-based stochastic modeling of the latent order book. Advised by Michael Benzaquen and Jean-Philippe Bouchaud.

Saildrone **Alameda, CA, USA**
Platform team intern **June - August 2019**
Processing of oceanographic data collected by drones, fault detection.

PUBLICATIONS

An updated list is available on [my Google Scholar page](#).

- Matthieu Blanke, Ronan Fablet, Marc Lelarge. Neural Incremental Data Assimilation. Accepted at the AI for Science workshop, at ICML 2024.
- Matthieu Blanke, Marc Lelarge. Interpretable Meta-Learning of Physical Systems. The Twelfth International Conference on Learning Representations, 2024 ([ICLR 2024](#)). Also presented at the Synergy of Scientific and Machine Learning Modeling workshop, at ICML 2023 (SynS & ML 2023).
- Matthieu Blanke, Marc Lelarge. FLEX: an Adaptive Exploration Algorithm for Nonlinear Systems. The Fortieth International Conference on Machine Learning, 2023 ([ICML 2023](#)). Also presented at the Machine Learning and the Physical Sciences workshop, at NeurIPS 2022.
- Matthieu Blanke, Marc Lelarge. Online greedy identification of linear dynamical systems. 61st Conference on Decision and Control, 2022 ([CDC 2022](#)).

Also presented at the Adaptive Experimental Design and Active Learning in the Real World workshop, at ICML 2022.

- Matthieu Blanke, Jose Moran, Pierre-Philippe Crépin, Jean-Philippe Bouchaud, Michael Benzaquen. Market impact in a multiple metaorder landscape. Under submission.

PROJECTS AND SOFTWARE

The projects are available on [my GitHub page](#).

Physics exercise book	Ongoing
Open source exercise book for undergraduate students preparing for the competitive exams for the top French engineering schools.	
Neural Incremental Data Assimilation	2024
Deep learning model for the data assimilation inverse problem, built with JAX.	
CAMEL	2023
Interpretable meta-learning of physical systems	
FLEX	2022
Adaptive exploration of physical systems.	
Deep Latent Variable Models	2021
Maximum likelihood sampling and missing data imputation based on deep learning. Supervised by S. Allassonnière.	
Market impact simulator	2020
Python module for noisy market impact experiments.	
Image segmentation: Random Walker and SegNet	2019
Theoretical study and Python implementation of the Random Walker algorithm. Performance benchmark versus the deep learning architecture SegNet. Supervised by S. Allassonnière.	
Automatic sport scene modeling	2020
A C++ library that automatically detects the players' positions on a sport video.	
Machine learning for power consumption forecast	2019
Clustering, pattern detection and statistics on weather data from Météo France to predict energy consumption. Implemented the algorithms from scratch in C++.	
IP-over-Discord	2018
A C program and a Node.JS Discord Bot for IP-over-Discord network tunneling.	

TALKS

02/2025	Columbia LEAP center, invited talk	New York
Interpretable Meta-Learning of Physical Systems.		
09/2024	Huawei Lavender Summit 2024, invited talk	Bordeaux
Interpretable Meta-Learning of Physical Systems.		
03/2024	Mines Paris Geosciences Department, invited talk	Paris
Interpretable Meta-Learning of Physical Systems.		
07/2023	MLIA seminar, invited talk	Paris
Interpretable Meta-Learning of Physical Systems.		
12/2023	ICML in Paris, contributed talk	Paris
FLEX: an Adaptive Exploration Algorithm for Nonlinear Systems.		

04/2023	Oral presentation for Safran Research Exploration of physical systems.	Paris
12/2022	ML4Physical Sciences workshop, poster presentation Online exploration of nonlinear physical systems.	Paris
10/2022	Inria PhD Seminar, oral presentation Exploration of physical systems.	Paris
06/2022	GdR IASIS, poster presentation Online greedy identification of physical systems. "Apprentissage et modélisation physique" workshop.	Paris
09/2021	CIRM Workshop, poster presentation Deep learning isochronism. "On Future Synergies for Stochastic and Learning Algorithms" workshop.	Marseille

TEACHING

Mines PSL Teaching assistant Probability theory. Differential equations.	Paris, France Fall 2023
Université Paris Cité Teaching assistant Numerical physics.	Paris, France Spring 2023
Université Paris 1 Panthéon-Sorbonne Teaching assistant Statistics.	Paris, France Fall 2022
Université Paris Cité Teaching assistant Numerical physics.	Paris, France Spring 2022
Université Paris 1 Panthéon-Sorbonne Teaching assistant Statistics.	Paris, France Fall 2021
École polytechnique, X-Talents Oral Examiner Weekly mathematics and physics oral tests preparing undergraduate students for the competitive entrance examinations of French top engineering schools.	Paris, France 2018

SERVICE

Organizer of Inria Argo team's seminar	2022-2024
Reviewing ICLR, ICML Workshop Synergy of Scientific and Machine Learning Modeling, NeurIPS workshop on Machine Learning and the Physical Sciences, IEEE Control Systems Letters, IEEE CDC	

SCHOLARSHIPS

2024	Google.org "AI for societal impact" scholarship
2021	École polytechnique's AMX PhD Scholarship

SKILLS

Computer languages Python, Julia, C, C++, Node.js, $\text{\LaTeX}$

Python frameworks PyTorch, JAX

LANGUAGES

Frenc Native

Italian Native

English Fluent

German Advanced

Spanish Conversational